FlowRecon-Bench: A benchmark for sparse-sensor flow-field reconstruction across flow regimes, with calibrated uncertainty

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Reconstructing a full flow field from sparse point sensors is a core task of experimental and operational fluid mechanics, and a rapidly growing machine-learning literature now addresses it with deterministic operator learners and conditional generative models. Yet the field has no shared benchmark: published methods are evaluated on incompatible, often temporally leaky splits, ensembles are drawn but never scored with proper probabilistic metrics, and classical baselines are rarely reported. We present FlowRecon-Bench, a benchmark spanning three flow regimes of increasing dimensionality and dynamical complexity — laminar cylinder vortex shedding (2D, multi-Reynolds), two-dimensional Kolmogorov turbulence, and three-dimensional isotropic turbulence (JHTDB) — evaluated with leakage-audited splits, frozen sensor manifests, per-channel accuracy, proper scoring rules (CRPS, coverage, spread–error), physics metrics (windowed spectra, structure functions), and matched cost accounting (training and inference runtime and peak memory) for every method. Across a fidelity-audited fleet of twelve-plus methods — interpolation and gappy-POD anchors, deterministic operator learners, latent and ambient generative models including DMF-Gen — we find that method ranking is regime-dependent. Gappy POD wins the laminar wake at 0.056 relative L_2, 66% below the best learned method; a deterministic operator learner takes 2D turbulence on relative L_2 (0.385) while ranking last among generative entries on CRPS; and on 3D turbulence no method leads every channel, with inverse-distance weighting matching the best learned observed-channel error. No method recovers unobserved channels of 3D turbulence, though every trained method recovers the unobserved pressure of the laminar wake; every scored ensemble is under-dispersed, and post-hoc recalibration partially repairs it. Benchmark scores are systematically worse than literature-reported numbers on the same data, and we show by controlled ablation that the gap is evaluation protocol — sensor layout, split design, and reporting convention — rather than model capability; the same checkpoint improves by 25% when its scattered sensors are rearranged onto a uniform lattice, while split leakage proves severe in the small-data 3D regime and negligible at 2D data scales. We release the datasets, splits, sensor manifests, configurations, and evaluation driver as a versioned artifact. [TODO: finalize abstract numbers and artifact DOI]

I. INTRODUCTION

Flow fields govern systems that experiments can only observe pointwise: particle-image velocimetry resolves planes of a turbulent flow while the surrounding volume goes unmeasured, flight-test campaigns record surface pressure at taps while the off-body flow that determines performance is inaccessible, and environmental monitoring networks sample the atmosphere at stations kilometers apart. Recovering the full field from such sparse, indirect data is a long-standing inverse problem,[1–3] and at realistic sensor densities it is severely ill-posed: many fields fit the data, and the missing information sits disproportionately at the small scales. A machine-learning literature now more than a decade deep addresses it with deterministic reconstruction networks,[4–7] and, more recently, with conditional generative models that sample the distribution of fields consistent with the observations rather than regressing to its mean.[8–13]

What that literature does not have is a shared evaluation. Community benchmarks for scientific machine learning — PDEBench, The Well, CFDBench, AirfRANS, and their successors — pose forward-surrogate or uniform-downsampling super-resolution tasks, not reconstruction from scattered point sensors; the Johns Hopkins Turbulence Database is a query service with no canonical protocol, and published sparse-reconstruction papers each cut their own incompatible extraction from it. Three consequences follow, and this paper documents all three. First, *splits leak*: turbulence snapshots are strongly autocorrelated in time, and the standard random-in-time split makes every test frame a near-duplicate of a training frame. On our 3D turbulence data, frame-to-frame correlation is 1.00 at unit separation; under a shuffled split the strongest latent generative baseline scores 0.14 relative L_2, and under an honest spatially disjoint split the same model scores 0.56 — a factor of four, larger than every method-to-method gap we measure (Sec. V F). Second, *ensembles are drawn but not scored*: generative reconstruction papers routinely show mean-and-standard-deviation maps, several claim calibrated uncertainty on that evidence alone, and, to our knowledge, none reports a proper scoring rule or a calibration diagnostic for 3D single-snapshot reconstruction from scattered sensors;[14–18] every scored precedent in adjacent settings is under-dispersed. Third, *classical baselines are absent*: training-free interpolation and gappy proper-orthogonal-decomposition floors are rarely reported, and we find one of them (inverse-distance weighting) beats every deep model's observed-channel error on 3D turbulence at 1% sensor density. A consequence worth stating plainly up front: every number and image in this paper will look worse than its counterparts in the literature, and Sec. V F shows by controlled ablation — same checkpoints, protocols swapped — that this gap is the protocol, not the models.

FlowRecon-Bench is our response: a benchmark for sparse-sensor flow reconstruction built to answer the question a practitioner actually has — *which method family should I use, in which flow regime, at what sensor budget, with what uncertainty and at what cost?* It spans three regimes forming a ladder in dimensionality and dynamical complexity (Sec. II B): laminar cylinder vortex shedding, where the dynamics are low-dimensional and classical linear methods should be near-optimal; two-dimensional Kolmogorov turbulence, the standard testbed of the 2D generative literature;[8,9] and three-dimensional isotropic turbulence under a spatially disjoint evaluation protocol, where sensors observe only two of four channels. The protocol carries over to multiphysics and unstructured-geometry regimes — corrupted measurement operators and surface-confined sensing on body-fitted meshes — which we defer to follow-up work rather than report here. Every method is evaluated under identical seeded sensor draws, identical augmentation, leakage-controlled splits with a strict tune/test separation, per-channel accuracy, fair CRPS, coverage and spread–error diagnostics, windowed spectral metrics, and — because model selection in practice is a cost–benefit decision — measured training and inference runtime and peak memory for every row.

Our contributions are the benchmark and the findings it forces. (i) A three-regime task suite with leakage-audited splits, frozen sensor manifests, and a fingerprinted evaluation driver, released as a versioned artifact together with per-method configurations and audit notes. (ii) A fidelity-audited fleet of twelve-plus methods spanning classical anchors, deterministic operator learners, latent generative and ambient generative routes, parameter-matched where the published configurations allow and budget-disclosed where they do not (Sec. III). (iii) Findings that generalize beyond any single table: method ranking is regime-dependent; no method recovers channels the sensors do not observe in 3D turbulence (a fleet-wide identifiability limit, not a per-method defect); accuracy and sample realism trade off through the sampler's operating point, so ranking generative models on mean-field error alone is misleading (Sec. V A); every scored ensemble is under-dispersed, and a post-hoc recalibration fitted on a disjoint tuning split repairs stated coverage (Sec. V C); and the computational axis separates grid methods, whose memory walls we measure, from point methods, whose costs are flat in resolution (Sec. V E).

TABLE I. The three regimes of FlowRecon-Bench. Each step of the ladder adds one source of difficulty; the split column names the leakage-controlled protocol (Sec. II C).

Regime	Dataset	Points	Channels	Split	Axis isolated
2D laminar periodic	Cylinder wake (new)	$2.4{\times}10^4$	u, v, p	cross-Re	low intrinsic dim.
2D chaotic	Kolmogorov 256[28]	$6.6{\times}10^4$	ω	cross-traj.	2D multiscale
3D chaotic	JHTDB isotropic[19]	$1.95{\times}10^6$	U_x, U_y, U_z, p	spatial	unobs. channels

II. BENCHMARK DESIGN

A. Task

Let $u : \Omega \to \mathbb{R}^C$ be a flow field drawn from a distribution over fields (snapshots of a turbulent flow, solutions over varying geometry). It is observed through a measurement operator returning readings $\mathcal{O} = \{(\boldsymbol{x}_j,\, c_j,\, y_j)\}_{j=1}^{M}$ with $y_j = u_{c_j}(\boldsymbol{x}_j) + \varepsilon_j$, where $\boldsymbol{x}_j$ is a sensor location, c_j indexes the observed quantity, and ε_j is noise. The task is single-snapshot reconstruction: estimate u — or, for probabilistic methods, sample an approximation to $p(u \mid \mathcal{O})$ — at all query locations, with no access to sensor time histories, initial conditions, or the governing solver. This covers sensors scattered arbitrarily or confined to a surface, observed quantities forming a subset of the target channels, and corrupted readings. Sequential assimilation and forward surrogacy are deliberately out of scope; they are different problems with different information budgets.

B. Datasets: three flow regimes

The three regimes form a ladder in both spatial dimension and dynamical complexity — 2D laminar and periodic, 2D chaotic, 3D chaotic — so that each step isolates one source of difficulty while holding the task fixed (Table I); preprocessing details and download/regeneration recipes ship with the artifact.

a. Cylinder wake (new). Two-dimensional incompressible flow past a circular cylinder, generated with OpenFOAM 9 (`pimpleFoam`, laminar) on a 12-block O-grid of 23,800 cells with boundary-layer grading (`checkMesh` clean; maximum non-orthogonality 43.9°). Nondimensionalized by the cylinder diameter and free-stream speed ($D = U_\infty = 1$; $\nu = 1/\mathrm{Re}$), at $\mathrm{Re} \in \{60, 80, 100, 150, 200, 250\}$. Each case runs a transient phase to $t = 120$ and a sampling phase to $t = 270$ written every 0.5 time units — 300 snapshots spanning roughly 25 shedding cycles at $\mathrm{Re} = 100$. Strouhal numbers of the generated cases are verified against established measurements[20,21] as the physics-level validation of the data. The measured values are $St = 0.140$, 0.160, 0.173, 0.187, 0.200 and 0.207 at $\mathrm{Re} = 60$, 80, 100, 150, 200 and 250, within the scatter of the published St–Re relation over this range; mean drag lies between 1.40 and 1.46, consistent with the 6% blockage of the domain, and the lift-coefficient amplitude increases monotonically with Re (0.16 to 0.88). The dataset ships in two exports: the native cell-center point cloud (with the wall-adjacent cell ring indexed for a surface-tap sensing variant) and a 400×200 uniform-grid interpolation over $x \in [-3, 17]$, $y \in [-5, 5]$ with a body mask, for grid-locked methods — the need for the second export is itself a capability observation. Cross-Re protocol: train on $\mathrm{Re} \in \{60, 100, 150, 200\}$, evaluate on the held-out $\mathrm{Re} \in \{80, 250\}$ (one interpolative, one mildly extrapolative), with temporally blocked splits inside the training cases. An observe-u-only variant probes cross-channel inference (v, p unobserved) in a regime where, unlike 3D turbulence, the phase of the shedding cycle makes the unobserved channels largely determined by the observed one.

b. Kolmogorov turbulence. The two-dimensional forced-turbulence vorticity dataset of Ref. 8, standard in the 2D generative-reconstruction literature: 40 independent trajectories of 320 frames at 256^2. We stride time by 4 (80 frames per trajectory, 3,200 total) and hold out entire trajectories: 32 for training, 8 for evaluation, assigned by a fixed seeded permutation. Cross-trajectory holdout is the 2D analogue of the spatial holdout below — reconstruction is scored on realizations of the flow the model has never seen at any time.

c. 3D isotropic turbulence. Four 125^3 cutouts at disjoint spatial locations of the JHTDB `isotropic1024coarse` DNS,[19] 50 snapshots each at a stride of 100 database steps, fields (U_x, U_y, U_z, p), per-field standardized. Training uses cutouts 1–3; all reported evaluation uses cutout 4, spatially disjoint from everything seen in training. Sensors observe U_x and U_z only; U_y and p are never observed, making this the identifiability probe of the suite.

C. Splits and leakage control

Every split in FlowRecon-Bench is chosen against the autocorrelation structure of its data, and the protocol is enforced by tooling rather than convention. Turbulence snapshots correlate at $r = 1.00$ frame-to-frame and $r = 0.67$ at separation 100; random-in-time splits therefore measure near-duplicate interpolation (Sec. V F quantifies the distortion). The suite uses spatial holdout (a disjoint cutout) for 3D turbulence, trajectory holdout for Kolmogorov, Reynolds-number holdout for the cylinder, and temporally blocked splits with decorrelation guard gaps wherever time enters. Model selection is separated from reporting: held-out snapshots are partitioned into a tuning half (odd indices) used for any checkpoint or hyperparameter selection and a test half (even indices) on which every reported number is frozen. The evaluation driver fingerprints the protocol — snapshot indices, sensor counts, and the index-sum of every seeded sensor draw — and aborts on mismatch, so no two methods can silently see different observations. Two incidents during construction illustrate why the tooling exists: an audit found one baseline's visual-evaluation path silently collapsing the third spatial axis, and the split machinery itself carried a one-frame boundary leak (an `int()` where floating-point arithmetic demanded rounding: at train fraction 0.8, $1 - 0.8$ evaluates below 0.2 and the first held-out-trajectory frame joined the training block). Both were caught by cross-checks the artifact now runs automatically.

D. Sensor protocols

The primary protocol draws sensors uniformly at random over the domain, per observed channel, at densities from 0.1% to 2% of the discretization (regime-dependent; 1% is the headline density), with counts and layouts redrawn per training step for amortized methods and frozen seeded draws at evaluation. Sensor layout is not incidental to difficulty: at fixed count, rearranging scattered sensors onto a uniform lattice changes reconstruction error by tens of percent (Sec. V F), so the layout is part of the protocol specification rather than an implementation detail. Two layout variants accompany the random-scatter default — a uniform lattice at matched count, and sensors confined to the cylinder wall ring — and channel subsets are specified per regime (Sec. II B). Distinguishing per-frame, per-trajectory, and globally fixed sensor masks matters for comparability across papers;[22] FlowRecon-Bench uses per-frame draws with frozen evaluation manifests. The evaluation harness also implements corrupted operators (Gaussian sensor noise, contiguous occlusion, channel dropout) applied through shared seeded code so that every method sees bit-identical corrupted observations; those results belong to the multiphysics regime and are deferred with it.

E. Metrics

a. Accuracy, per channel. Relative L_2 per channel is primary, reported separately for observed and unobserved channels — aggregates average two different questions and are shown flagged, never headlined. For ensemble methods both the ensemble-mean error and the single-sample error are reported; the gap between them is the smoothing an ensemble average buys, and single-sample error measures whether individual realizations are physical.

b. Probabilistic scores. For an ensemble $\{s_k\}_{k=1}^K$ and truth y at a point, the fair CRPS estimator[14] is CRPS $= \frac{1}{K}\sum_k |s_k - y| - \frac{1}{2K(K-1)}\sum_{k\neq l}|s_k - s_l|$, averaged over points and fields in standardized units. The spread–error ratio is the ratio of ensemble standard deviation (RMS over points) to the RMS error of the ensemble mean (ideal 1.0); coverage$_{90}$ is the fraction of points whose truth falls in the central 90% ensemble interval (ideal 0.90); rank histograms accumulate the rank of the truth within the ordered ensemble. Deterministic methods enter the CRPS column exactly (CRPS reduces to MAE for a point forecast); their dispersion entries are undefined, not zero. Calibration metrics move with the operating point — sampler steps, ensemble size, sensor density — by more than they separate methods, so every calibration number carries its operating point and cross-method comparisons are made only at matched settings. Ensemble size deserves particular care: in a K-sweep on Kolmogorov (K=8–64, same frames and sensors), DMF-Gen's ensemble-mean rel-L_2 moved only 0.487→0.475 and CRPS was flat (the fair estimator is unbiased), but coverage$_{90}$ rose 0.59→0.73 — finite-ensemble quantiles are systematically too narrow, so K=8 *understates* every generative method's calibration. We therefore report accuracy and CRPS at K=8 and treat K=8 coverage as a conservative bound, with the sweep in the artifact.

c. Physics metrics. Shell-averaged energy spectra of single samples (never ensemble means, which destroy small scales by construction), band ratios against the reference solution, structure functions, and velocity-gradient PDFs. One estimator detail is load-bearing enough to state in the protocol: non-periodic cutouts must be windowed before the FFT. On the 3D data the raw spectrum of the *truth* falls only 2.4× from $k = 32$ to the Nyquist wavenumber while a variance-preserving Hann taper restores a 5.8×10^5 decay; without the taper, a smooth Gaussian-process draw with zero energy at those wavenumbers registers the same broadband floor as the DNS, and spectral ratios computed there are ratios of two leakage floors that tend to unity for every method. All spectral numbers in FlowRecon-Bench use the windowed estimator.

d. Cost. Model selection is a cost–benefit decision, so cost is a first-class metric family, measured under the same harness for every method on every dataset: training seconds per optimizer step, total GPU-hours to the finished model, peak training memory, inference wall-clock per full-field reconstruction, and peak inference memory. Rows without cost entries are treated as incomplete by the table-assembly tooling. A separate resolution-scaling study (Sec. V E) measures memory versus output discretization from 64^3 to 1024^3, with out-of-memory walls reported as observed failures, not extrapolations.

F. Fairness methodology

Benchmark tables are only as credible as their weakest baseline, so the fleet is engineered adversarially against our own biases. (i) *Upstream fidelity audits*: every baseline implementation is diffed against its published code before its numbers are admitted, and every declared adaptation (dimensionality ports, set encoders, sampler retuning) is documented in the per-method audit notes; audit incidents (a conditioning defect in the latent flow-matching baseline, sampler-parameter sensitivity in S3GM) were repaired or labelled before reporting. (ii) *Parameter matching*: learned methods are matched to ∼6.5M parameters where the published configuration allows; declared waivers (CoNFiLD's published prior at 118.9M) are printed in the table. (iii) *Identical conditions*: the same data files, splits, augmentation, normalization statistics, seeded sensor draws, and metric estimator for every row. (iv) *Budget disclosure*: optimizer steps and wall-clock are reported as run, not silently equalized, and the table notes which budgets exceed which. (v) *Tuning-budget disclosure*: the method we co-developed (DMF-Gen) received more configuration search than any baseline; upstream-faithful configurations are frozen as the headline rows and every tuned variant is a separately labelled arm. (vi) *Selection hygiene*: checkpoint selection uses the tuning split only (Sec. II C).

FIG. 1. Per-method capability profile across the deployment axes measured in this paper (filled = yes, half-filled = partial or adapted, open = no). "Dissipation-preserving" records the windowed dissipation-band ratios (Sec. V D); "memory flat vs resolution" and "sub-second inference" record the measured scaling study (Sec. V E); "recalibrable" records whether the post-hoc wrapper of Sec. V C applies. Axes are marked "no" only when the failure is structural.

III. METHODS EVALUATED

The fleet spans four families; Table II records which methods can enter which regimes, itself a benchmark result, and Fig. 1 expands it to the per-method deployment axes — representation freedom, uncertainty, spectral fidelity, and scaling — that Sec. V measures. No method fills every column: the ambient point-cloud route alone clears the representation and dissipation-band axes but is the slowest sampler; latent FM inverts that profile exactly. Per-method configurations, adaptations, and audit notes ship with the artifact.

a. Classical anchors. Four training-free floors run on every regime: nearest-neighbor interpolation (KD-tree), inverse-distance weighting over the $k = 8$ nearest sensors, gappy proper orthogonal decomposition[23] (POD basis from the training set, modal coefficients by least squares on the sensor readings, rank 80), and the training-mean constant. Distance handling is non-periodic by default — an earlier periodic implementation understated these floors on non-periodic cutouts and was corrected in audit — with periodic variants where the domain genuinely is periodic. As deterministic point forecasts their CRPS equals their MAE. These anchors define what any learned method must beat to justify its training cost.

b. Deterministic operator learners. Senseiver[5] is an attention-based point-native reconstructor mapping arbitrary sensor sets to query values through a latent bottleneck; it trains under the identical sensor randomization as the amortized generative methods. FNO3D places a spectral (FFT-grid) backbone inside the same conditional-generation framework as DMF-Gen, isolating the backbone axis from the formulation axis. DeepONet[24] enters in upstream-faithful form (branch/trunk with a declared set-encoder adaptation for

TABLE II. Capability matrix: which method families can honestly enter which regimes. RS = requires resampling to a grid, lossy near boundaries — the cylinder's body-fitted mesh is the case in point, and the reason the dataset ships a second, interpolated export (Sec. II B).

Family	Cyl.	Kolm.	JHU
Interpolation/POD anchors	✓	✓	✓
Point-attention det. (Senseiver)	✓	✓	✓
Grid det. (FNO-class)	RS	✓	✓
DeepONet-class	✓	✓	✓
Latent-grid generative (LFM)	RS	✓	✓
Latent neural-field gen. (CoNFiLD)	✓	✓	✓
Voxel/video diffusion (Gen4Turb, S3GM)	RS	✓	✓
Point-token generative (SiT-point)	✓	✓	✓
Ambient point-cloud FM (DMF-Gen)	✓	✓	✓

variable sensor sets); its near-constant predictions are the pre-registered structural consequence of global mean pooling in the branch, and a structured-branch redesign ("DeepONet++", multiresolution binned pooling) is reported as a labelled improvement arm. Deterministic models return conditional-mean estimates: optimal in squared error, systematically smoothed, and structurally unable to report what the sensors leave undetermined.

c. Latent generative. Latent flow matching (LFM) is the latent-diffusion-family stand-in:[11,25] a convolutional autoencoder trained on full fields, followed by conditional rectified flow in its latent space with a point-encoder conditioning branch. It is the strongest aggregate performer on 3D turbulence and carries the family's structural signature, a dissipation-range spectral floor inherited from the autoencoder.[26,27] CoNFiLD[11] is the published latent neural-field competitor: a conditional-neural-field autodecoder compressing each snapshot to a latent vector, an unconditional diffusion prior over latents, and diffusion posterior sampling for conditioning; three arms are reported (published prior; capacity-matched; 384-dimensional), with the latent-capacity ceiling of the autodecoder measured explicitly as the decoder bound.

d. Ambient generative. Gen4Turb[12] is voxel diffusion: an elucidated-diffusion 3D U-Net with mask-channel conditioning, run at its native 120^3 on the 3D data and retrained under the benchmark's observation model (the most generous variant is reported). S3GM[9] is a pretrained video-diffusion prior with sampling-time guidance, natively two-dimensional; its 3D entry is a declared adaptation treating one spatial axis as video time, and its published fixed guidance weights required tuning-split renormalization to converge at 3D scale. SiT-point adapts a scalable interpolant transformer[28] with a point tokenizer (not patchification): sensors and queries are tokenized directly, generation runs in fixed-size token chunks, and the chunk budget is the visible constraint (member-level chunk-incoherent speckle that ensemble averaging hides — one reason FlowRecon-Bench reports member-level metrics). DMF-Gen[29] performs conditional rectified flow directly on field values at scattered points — ambient function space, no autoencoder, no voxelization — with a Gaussian-process source shared across discretizations, conditioning factored into a global cross-attention summary plus a local k-nearest-sensor retrieval per query, a Monte-Carlo subsampled training objective (a random $\sim$2% of points per step), and training amortized over sensor counts, layouts, noise, and observed-channel subsets. We evaluate it as published, at the benchmark's parameter budget; it is one row of the fleet, not the paper's contribution.

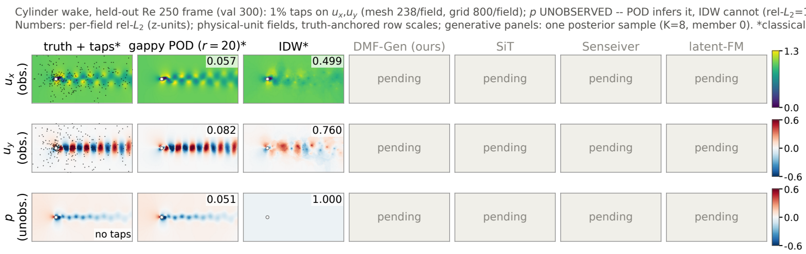

FIG. 2. Cylinder wake, held-out Re=250, single reconstructions from 1% taps on (u, v) under identical sensor sets. The pressure row is the regime's signature: gappy POD infers the never-observed channel from modal structure, while interpolation is blind to it by construction. Per-panel numbers are per-field relative L_2 in standardized units.

IV. RESULTS BY REGIME

A. Laminar vortex shedding: where classical methods should suffice

The dataset was generated for this benchmark (Sec. II B) and verified against the literature before any model saw it: Strouhal numbers 0.140/0.160/0.173/0.187/0.200/0.207 for Re 60–250 track the Williamson–Norberg correlations, mean drag sits where the 6% blockage predicts, and lift amplitude grows monotonically with Re. Sensors observe (u, v) at 1% of the 23,800 mesh cells (238 per field); pressure is never observed; evaluation is on the held-out Reynolds numbers $\{80, 250\}$.

The classical floors decide this regime before the learned fleet reports. The training family's POD spectrum is essentially exhausted at rank 20 (99.7% energy), and *gappy POD with 20 modes reconstructs held-out Reynolds numbers at 0.056 relative L_2 — including the never-observed pressure at 0.05* — because the modal coefficients inferred from sparse velocities carry the full coupled state. Pure interpolation cannot do this (p column: 1.0 by construction) and trails badly on the velocities as well (IDW 0.71 aggregate). This is the mirror image of 3D turbulence, where the same POD basis was useless (0.95, Table V) and interpolation was the strong floor: *which classical method is the one to beat is itself regime-dependent*, and a learned method only earns its complexity here if it approaches the POD floor at comparable cost.

No method in the fleet does (Table III). GeoFNO leads the learned entries at 0.093 aggregate rel-L_2, 66% above the rank-20 POD floor of 0.056, and the ordering behind it spans a factor of four before S3GM diverges. DMF-Gen places fourth at 0.221, behind latent FM by 0.030 (95% CI $[0.014, 0.045]$) and behind GeoFNO by 0.128 (95% CI $[0.123, 0.134]$, 0 of 50 paired snapshots). Reporting this plainly matters more than the individual ordering: a benchmark whose regimes all favour the same family would not be measuring regime dependence.

Two results survive the POD comparison. Every learned method infers the never-observed pressure (0.088 to 0.274) while both interpolants return the training mean there by construction, so the cross-channel inference that fails in 3D turbulence (Sec. V B) succeeds here for every trained model. And the calibration ordering inverts relative to JHU: latent FM reaches spread–error 1.26 and coverage$_{90}$ 0.89 on the cylinder against DMF-Gen's 0.57 and 0.51, the reverse of the 3D result where DMF-Gen was the better-dispersed of the two. Posterior spread that is well scaled in one regime is not well scaled in another, which is an argument for the post-hoc recalibration of Sec. V C rather than for either architecture.

Held-out Reynolds numbers separate interpolation from extrapolation. Every learned method scores better at Re=80, inside the training range $\{60, 100, 150, 200\}$, than at Re=250 outside it: GeoFNO 0.074 versus 0.111, DMF-Gen 0.192 versus 0.250, latent FM 0.168 versus 0.215, Senseiver 0.129 versus 0.162. The degradation is 26–50% and no method is exempt.

TABLE III. Cylinder wake, Reynolds-number holdout ($\{80, 250\}$ held out from $\{60, 100, 150, 200\}$; 50 frames stratified across both; (u, v) observed at 1% of 23,800 mesh cells, p never observed; identical canonical seeded draws). Starred column is the unobserved channel. Inference cost per field on one H100; classical methods run on CPU.

Method	u	v	p^*	agg	CRPS	Cov.$_{90}$	s/field	GB
Gappy POD (r=20)	**0.043**	**0.075**	0.051	**0.056**	**0.029**	—	—	—
Gappy POD (r=80)	0.055	0.103	**0.044**	0.067	**0.029**	—	—	—
GeoFNO	0.087	0.103	0.088	0.093	0.053	—	0.014	0.34
Senseiver	0.101	0.240	0.095	0.145	0.080	—	0.009	0.29
Latent FM (NFE 4)	0.170	0.239	0.166	0.192	0.076	**0.89**	0.029	**0.21**
DMF-Gen (NFE 4)	0.182	0.353	0.128	0.221	0.098	0.51	0.080	0.31
SiT (50 steps)	0.302	0.409	0.274	0.328	0.106	0.55	0.187	0.28
MLP–RBF	0.265	0.679	0.221	0.388	0.213	—	**0.006**	0.25
IDW (k=8)	0.505	0.619	1.000	0.708	0.440	—	—	—
Nearest neighbor	0.512	0.614	1.000	0.709	0.422	—	—	—
S3GM (200 steps)†	1.035	1.017	1.005	1.019	0.521	0.58	29.2	20.1
Training mean	1.000	1.000	1.000	1.000	0.664	—	—	—

†S3GM's guided sampler does not converge on this dataset and scores below the training-mean floor; its Kolmogorov run is healthy (Table IV). Reported rather than withheld, with the sampler configuration in the artifact.

B. Two-dimensional Kolmogorov turbulence

Sensors observe vorticity at 1% of the 256^2 points (655 sensors); learned models are scored on 50 evenly spaced frames of the eight held-out trajectories (K=8, identical canonical seeded draws across methods; classical floors on all 640). Table IV reports the fleet against the floors, and Fig. 5 (left) the sensor-count sweep.

The accuracy and probabilistic rankings dissociate. GeoFNO takes the rel-L_2 column at 0.385, ahead of DMF-Gen by 0.103 (95% CI $[0.094, 0.112]$, better on 50 of 50 paired snapshots), and ranks last among the four generative entries on CRPS at 0.267, behind DMF-Gen by 0.032 (95% CI $[0.026, 0.037]$, 46 of 50). A deterministic regressor minimizes squared error by predicting the conditional mean, which is what rel-L_2 rewards; the proper score charges it for having no distribution. Both orderings are significant on the same 50 snapshots, so a paper reporting either metric alone would name a different winner (Sec. V A). DMF-Gen and SiT are indistinguishable (ΔCRPS 0.001, 95% CI $[-0.002, 0.004]$; Δrel-L_2 -0.006, 95% CI $[-0.012, 0.000]$).

The 3D ranking does not transfer. Latent FM, which wins the JHU aggregate, scores 0.600 here at the matched ~50k-step budget, below the inverse-distance floor of 0.546, and DMF-Gen beats it on both metrics on 50 of 50 snapshots. Senseiver reaches 0.913, near the training-mean floor, against a final training loss of 0.166 and validation loss of 0.912. S3GM buys the best CRPS (0.231) and the only coverage above 0.75 for 22.1 s and 17.4 GB per field, three orders of magnitude more inference cost than latent FM. Gappy POD's rank-80 basis captures 70.5% of training energy and the method reaches only 0.680: two-dimensional turbulence is not low-rank in the POD sense, and inverse-distance weighting beats it by 20%. [TODO: 2D spectra panel.]

C. Three-dimensional isotropic turbulence: the full fleet

The 3D turbulence benchmark is the controlled comparison: a periodic-topology regular grid is native terrain for voxel, patchified, and latent-grid methods, so every family competes at full strength. Sensors observe U_x and U_z at 1% of 1.95×10^6 points per channel; U_y and p are never observed. Table V reports per-channel accuracy and probabilistic scores for the

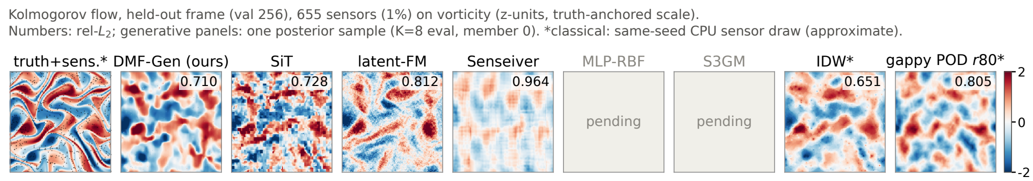

FIG. 3. Kolmogorov flow, held-out trajectory, single reconstructions of vorticity from 655 scattered sensors (1%), identical sensor set for every panel. Architectural signatures are directly legible: the patch tokenizer's block structure, the latent decoder's uniform high-frequency texture, the deterministic learner's smoothing, and the interpolants' sensor-centred artifacts.

TABLE IV. Kolmogorov 256^2, trajectory-holdout evaluation (vorticity observed at 1% = 655 sensors, identical canonical seeded draws; learned rows: 50 evenly spaced held-out frames, K=8; classical floors: all 640). Inference cost per field on one H100. Deterministic methods have equal mean and sample columns and no dispersion entries.

	Rel. L_2					
Method	mean	sample	CRPS	Cov.$_{90}$	s/field	GB
GeoFNO	**0.385**	0.385	0.267	—	0.008	0.36
DMF-Gen (NFE 4)	0.487	**0.578**	0.235	0.59	0.19	0.76
SiT (50 steps)	0.494	0.590	0.234	0.64	0.18	0.24
S3GM (200 steps)	0.498	0.700	**0.231**	**0.80**	22.1	17.4
IDW (k=8, periodic)	0.531	0.531	0.375	—	—	—
IDW (k=8, non-per.)	0.546	0.546	0.385	—	—	—
Nearest neighbor (per.)	0.582	0.582	0.371	—	—	—
Latent FM (NFE 4)	0.600	0.731	0.310	0.64	**0.03**	**0.19**
MLP–RBF	0.639	0.639	0.469	—	0.009	0.79
Gappy POD (r=80)	0.680	0.680	0.511	—	—	—
Senseiver	0.913	0.913	0.697	—	0.011	0.71
Training mean	1.000	1.000	0.753	—	—	—

full fleet on all 50 held-out snapshots of the spatially disjoint cutout under canonical seeded draws; Table VI reports the matched cost accounting.

The honest headline is that no method leads every column. SiT-point posts the best observed-channel errors (0.143/0.132) at roughly 150 s per field — 239 sequential 8,192-token chunks at 32 steps — and its single samples carry chunk-incoherent speckle that the ensemble mean averages away. Latent flow matching leads the aggregate, the unobserved channels, and CRPS: its convolutional decoder emits globally structured fields even where no sensor constrains them, at the dissipation-range price quantified in Sec. V D. Non-periodic inverse-distance weighting — a training-free interpolant — edges the best learned observed-channel error (0.174 vs. 0.176 on U_x for DMF-Gen), so no learned method beats interpolation on observed channels at 1% density; the learned and generative payoffs concentrate where the problem is ill-posed — unobserved channels, sparser sensing — and in sampling cost (DMF-Gen reaches its accuracy in 4 network evaluations against 32–1000 for the guided-diffusion and transformer baselines). DeepONet's near-constant predictions are the structural consequence of global mean pooling in its branch; the labelled DeepONet++ arm (structured pooling) improves U_x from 0.593 to 0.512 and remains behind the point-attention family.

Per-snapshot paired tests on the same 50 snapshots separate the differences that survive snapshot variance from those that do not. Against latent flow matching, DMF-Gen is behind on rel-L_2 by 0.029 (95% CI $[0.009, 0.049]$, better on 16 of 50) and ahead on CRPS by 0.011 (95% CI $[-0.003, 0.025]$, better on 30 of 50, not significant); a two one-sided test bounds the CRPS difference within ± 0.03 at $p = 0.004$, so the two are equivalent on the

TABLE V. 3D isotropic turbulence, spatially disjoint evaluation region (1.95M points; U_x, U_z observed at 1% per channel, U_y^*, p^* unobserved; canonical seeded sensor draws, all n=50 held-out snapshots, K=8 ensembles at each method's listed sampler setting). Per-channel relative L_2 of the ensemble mean is primary; agg† averages all four channels and is flagged, not headlined. Unobserved-channel error ≥ 1.0 is worse than predicting the training mean; values near 1.0 across the fleet reflect a shared identifiability limit of point-conditioned methods (Sec. V B). Spread–error and coverage ideals are 1.0 and 0.90; for deterministic methods CRPS reduces to MAE and dispersion is undefined. Step budgets are as-run, not equalized (Table VI). Labelled improvement arms (Senseiver capacity, DeepONet++, CoNFiLD repairs, S3GM normalized guidance) are reported in the artifact, including the CoNFiLD strict-capacity arm (rel-L_2 0.90 at 2048 latent dimensions), which does not change the ordering.

Method	U_x	U_y^*	U_z	p^*	agg†	CRPS	Spr/Err	Cov.$_{90}$
DMF-Gen (NFE 4)	0.176	1.048	0.182	0.964	0.593	0.291	0.63	0.54
Latent flow matching	0.248	**0.727**	0.257	**0.646**	**0.469**	**0.244**	0.68	0.56
SiT-point (N=32)	**0.143**	0.968	**0.132**	0.937	0.545	0.338	0.75	0.58
FNO3D (same framework, NFE 4)	0.188	0.904	0.191	1.061	0.586	0.269	**0.94**	**0.63**
CoNFiLD (published prior, DPS 1000)	0.409	0.751	0.374	1.007	0.635	0.371	0.82	0.57
CoNFiLD (capacity-matched)	0.488	1.089	0.420	1.487	0.871	0.562	0.62	0.45
CoNFiLD (384-d)	0.524	0.604	0.451	1.401	0.745	0.480	0.69	0.50
Gen4Turb (anneal, 32-step)	0.224	0.996	0.455	1.005	0.670	0.407	0.37	0.33
S3GM (JHU-tuned, N=200)	sampler diverges at 3D scale (rel-$L_2 \sim 10^6$); see text							
Senseiver (deterministic)	0.361	1.077	0.380	0.978	0.699	0.479	—	—
DeepONet (deterministic)	0.593	0.972	0.652	0.985	0.800	0.570	—	—
Nearest neighbor (KD-tree)	0.209	1.000	0.219	1.000	0.607	0.406	—	—
IDW (k=8, non-periodic)	0.174	1.000	0.181	1.000	0.589	0.396	—	—
Gappy POD (r=80)	0.523	1.306	0.540	1.443	0.953	0.673	—	—
Training mean	1.000	1.000	1.000	1.000	1.000	0.786	—	—

proper score at that margin while the rel-L_2 deficit is real. Against Senseiver both metrics favour DMF-Gen decisively (rel-L_2 -0.129, 95% CI $[-0.161, -0.097]$; CRPS -0.251, 95% CI $[-0.270, -0.232]$, 50 of 50). Snapshot-to-snapshot variance exceeds most method-to-method differences on this dataset, so aggregate columns alone cannot establish an ordering. [TODO: seed replicates for the DMF-Gen row.]

V. CROSS-CUTTING ANALYSES

A. The distortion–perception tradeoff, operationally

Relative L_2 of the ensemble mean is minimized by the posterior mean — by construction it rewards collapse toward a smooth conditional average — while proper scores and physics metrics reward calibrated diversity and sharp members. The benchmark measures the tension directly through the sampler's operating point: for the rectified-flow methods, two integration steps give the best mean-field error and the sharpest single samples, four steps are the CRPS optimum, and dispersion rises monotonically with further steps at flat CRPS (spread–error 0.65 at NFE 2 to 0.92 at NFE 16 for the $\lambda_{\rm sp}$=0 DMF-Gen variant). A spectral training penalty moves along the same frontier: it buys single-sample sharpness and inertial-band fidelity at a measured cost in dispersion. The practical rule the benchmark enforces: never rank generative reconstruction methods on mean-field error alone, and attach the sampler operating point to every probabilistic number.

TABLE VI. Cost accounting for the 3D-turbulence fleet (H100-SXM 80 GB). Steps are as-run, not equalized: every external baseline received at least DMF-Gen's 48k optimizer steps except SiT-point (wall-clock-matched, 38.5k); CoNFiLD stages train on a wall-clock budget. "(d)" = derived from evaluation-job wall time rather than a dedicated benchmark. Derived timings include per-call setup that a dedicated benchmark amortizes: DMF-Gen measures 17 s per field on the evaluation path, which recomputes the sensor-conditioning branch at every solver step, against 2.9 s under the benchmark harness that caches it across steps (both at NFE 4, same hardware). Derived and benchmarked timings are therefore not interchangeable, and rows carrying "(d)" are upper bounds.

Method	Params	Opt. steps	Train s/step	Train GPU-h	Train peak GB	Infer s/field
DMF-Gen (NFE 4)	6.51M	48k	0.574	7.7	53.6	∼17 (d)
Latent flow matching	6.68M	209k+95k	0.238	13.9	3.6	0.019
SiT-point (N=32)	6.57M	38.5k	1.81	19.4	26.8	∼150 (d)
FNO3D (NFE 4)	6.73M	79.9k	0.766	17.0	48.6	0.097
CoNFiLD (pub. prior)	118.9M^{w}	wall	35.5/ep	∼19	4.7	∼130 (d)
Gen4Turb (anneal)	6.16M	105k	0.598	17.4	47.4	1.65
S3GM (JHU-tuned)	6.35M	105k	0.635	18.5	26.3	∼46
Senseiver	6.42M	79.2k	0.776	17.0	22.4	<1
DeepONet	6.51M	82.0k	0.29	6.7	15.7	0.23
IDW k=8 (CPU)	—	—	—	—	10 (RSS)	0.68
Gappy POD r=80 (CPU)	—	45 s fit	—	—	10 (RSS)	0.14

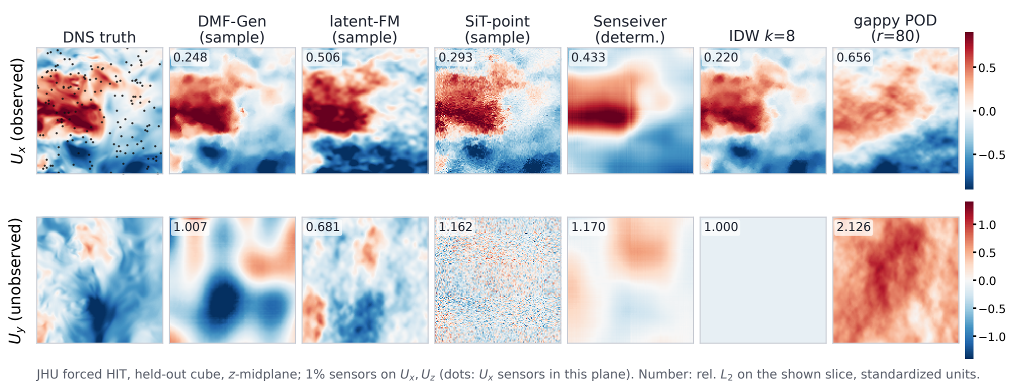

FIG. 4. 3D turbulence, held-out cutout, z-midplane of single reconstructions from 1% sensors on U_x, U_z (identical sensor set throughout; dots mark in-plane sensors). Top row: an observed channel, where every route is competitive. Bottom row: the unobserved U_y, where the fleet-wide identifiability limit is visible — interpolation returns the training mean, gappy POD extrapolates away from the truth, the point-token sampler emits chunk-incoherent noise, and only decoders imposing global field structure produce plausible (if uncorrelated) fields.

B. Unobserved channels: a fleet-wide identifiability limit

On 3D turbulence, no method in the fleet reconstructs the unobserved U_y below 0.60 relative L_2, and most sit near or above 1.0 — worse than predicting the training mean (Table V). The two partial exceptions (latent FM at 0.727, CoNFiLD-384d at 0.604) are the two methods whose decoders impose the strongest global field structure, and both pay for it elsewhere (dissipation-range truncation; observed-channel accuracy). This is not a per-method defect: at 1% point sensing on two velocity components, the instantaneous realization of a third component of 3D turbulence is close to unidentifiable, and only its statistics are recoverable. The observation disciplines how such results should be reported: per channel, with unobserved channels flagged.

The cylinder supplies the contrasting case. Its pressure field is never observed and every

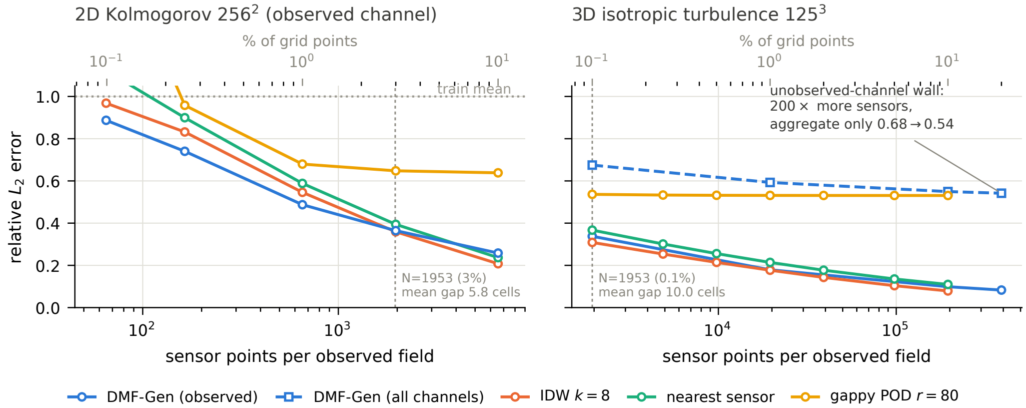

FIG. 5. Reconstruction error versus the *number* of sensor points, in 2D (left, Kolmogorov 256^2) and 3D (right, JHU 125^3 cross-cube; solid = observed-channel rel-L_2, dashed = aggregate over all four channels). The two panels sit on opposite sides of the density story: the same sensor count that leaves a 5.8-cell mean gap in 2D leaves a 10-cell gap in 3D, and while every observed-channel error falls steadily with sensor count — DMF-Gen tracking the interpolation slope from 0.33 to 0.09 over 200× — the 3D *aggregate* saturates near 0.54 because the unobserved channels do not identify (Sec. V B). Gappy POD is flat at every density: 3D isotropic turbulence has no low-rank structure for added sensors to project onto. DMF-Gen curves: K=8, NFE 4, 50 held-out snapshots; classical curves: same seeded sensor draws. Only DMF-Gen was swept across sensor count on the 2D panel; the remaining methods were trained at the single 1% budget of Table IV.

trained model recovers it, from GeoFNO at 0.088 to SiT at 0.274 (Table III), because the shedding phase inferred from velocity taps determines the rest of the state. The distinction is not method capability but whether the observations identify the unobserved channel at all: at 1% sensing on two velocity components of 3D turbulence they do not, and no architecture in the fleet overcomes that. Reporting a single aggregate error over observed and unobserved channels averages these two situations together and hides both.

C. Calibration: characterized, then repaired

Every scored ensemble in the fleet is under-dispersed at the headline operating point (spread–error 0.37–0.94 against the ideal 1.0; coverage 0.33–0.63 against 0.90), consistent with every scored precedent in adjacent literatures.[15,16] For the best-characterized model the miscalibration has a low-dimensional mechanism: predictive spread follows a distance-to-nearest-sensor profile with a hard floor (≈0.09 in standardized units), the profile is density-invariant, and the sampler step count rescales it wholesale (≈1.5× from NFE 4 to 16). That structure makes post-hoc repair tractable, and the benchmark ships two estimators fitted strictly on the tuning split and frozen on test: a per-density scalar spread multiplier, and conformalized quantile scaling binned by distance-to-nearest-sensor, which carries a finite-sample marginal coverage guarantee on exchangeable points. The identical wrapper is offered to the generative baselines, so the repaired comparison is not one-sided. Three honesty requirements govern the reporting: the repair is post hoc and stated as such; it rescales stated uncertainty and cannot sharpen the spread floor, which remains the residual limitation; and all repaired numbers carry their operating point. [TODO: TEST-split before/after spread–error and coverage for both estimators, DMF-Gen and latent FM, at 2–3 densities (fits exist: `src/recalibrate_spread.py`, `src/conformal_recalib.py`; per-point dumps queued).]

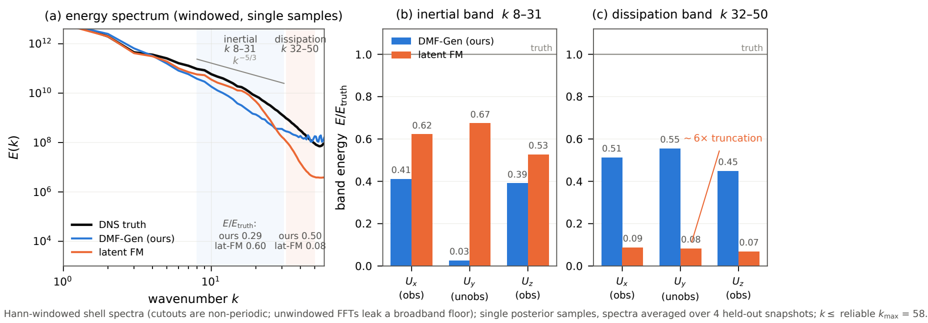

FIG. 6. Turbulence statistics of single posterior samples on the held-out 3D region (4 snapshots, NFE 4, Hann-windowed; ensemble means are deliberately not shown — averaging destroys small scales by construction). Left: shell-averaged energy spectra — the latent baseline tracks the DNS through the inertial range then falls away sharply in the dissipation band (autoencoder floor); the ambient model sits lower in the inertial range and flattens into a noise plateau at high k. Middle: longitudinal structure functions. Right: velocity-gradient PDFs.

D. Spectral fidelity splits by band

Windowed shell spectra of single samples (Fig. 6) separate the generative families where pointwise error cannot. On 3D turbulence the latent route holds more inertial-range energy than the ambient point-cloud route (0.62 vs. 0.41 of DNS energy on the observed U_x, $k \in [8, 31]$) but collapses in the dissipation band, retaining 0.07–0.09 of DNS energy over $k \in [32, 45]$ against the ambient model's 0.40–0.46 — the autoencoder spectral floor[26,27] appearing where it was predicted, a factor-of-five separation invisible to relative L_2. Neither family reproduces the DNS spectrum from 1% sensors: the ambient model's high-k content flattens into a broadband-noise plateau on the unobserved pressure channel (3.5–4.8× DNS energy), and on channels no sensor constrains its samples stay near its spectrally empty source prior (inertial ratio 0.03–0.06 on U_y, versus 0.67 for the latent decoder that paints texture everywhere). Member-level spectra, windowed estimation, and per-channel reporting are all load-bearing; ensemble-mean spectra and unwindowed estimators each silently manufacture different conclusions.

E. Computational cost and scaling

Cost separates the fleet along two axes. At the benchmark resolution (Table VI), total training cost to a finished model spans 7–19 GPU-hours across learned methods — comparable — but inference spans five orders of magnitude, from 19 ms (latent decode) to minutes (chunked token generation, DPS optimization), and peak training memory spans 3.6–54 GB. Against resolution (Fig. 7), the fleet splits into two families: grid computation (voxel diffusion, convolutional autoencoders) scales cubically with measured out-of-memory walls at 320^3 (training, batch 1) and 640^3, while point computation (attention over sensors with chunked queries) is flat to 1024^3 in both training and inference memory. The honest converse is that grid computation is cheaper below the crossover ($\sim 240^3$ for inference: a convolutional decode of a 125^3 field costs 6 ms where chunked point evaluation costs seconds). No family dominates; the benchmark's contribution is that the walls and crossovers are measured, so a practitioner can place their discretization on the axis before choosing a method family.

Figure 8 makes the same point from the accuracy side: on the cost–accuracy plane alone the latent baseline dominates every method including DMF-Gen, and the case for ambient point-cloud generation rests entirely on the axes the plane omits — resolution scaling, discretization freedom, and dissipation-range fidelity.

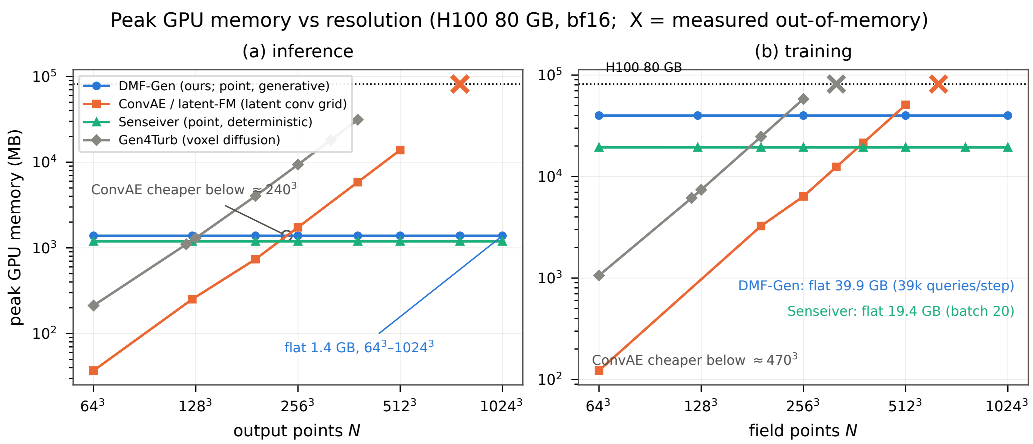

FIG. 7. Measured peak memory versus output resolution (H100, bf16, log–log; OOM markers are measured failures, not extrapolations) for full-field sampling (left) and training steps (right). Grid methods (voxel diffusion, convolutional AE) scale cubically and fail at 320^3 and 640^3; point methods (DMF-Gen, Senseiver) are flat to 1024^3 (10^9 points). Annotation: fraction of the field supervised per training step — 100% forced for grid methods, a fixed 39k-point budget for the subsampled point method.

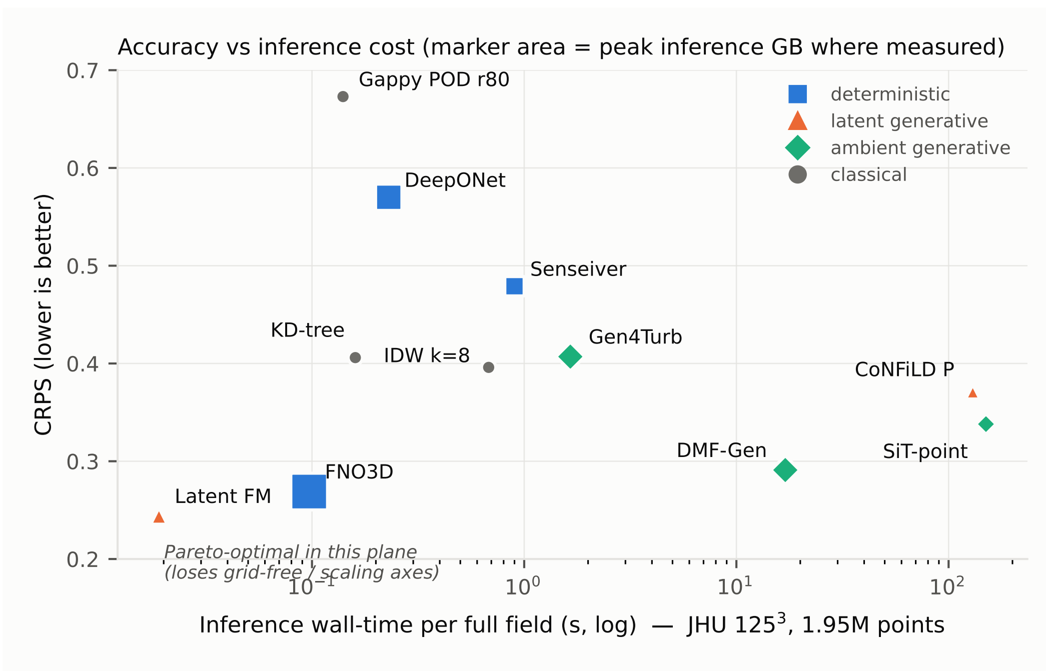

FIG. 8. Probabilistic accuracy (CRPS, JHU cross-cube, n=50) against inference wall-time per full 125^3 field, marker area proportional to peak inference memory where measured (Table VI). At the benchmark resolution the latent flow-matching baseline Pareto-dominates this plane outright — fastest and lowest CRPS — which is precisely why a single cost–accuracy plot is an insufficient basis for method choice: the axes it cannot show (Fig. 1; the scaling walls of Fig. 7; the dissipation-band ratios of Sec. V D) are where the grid-locked methods lose. S3GM is absent: its upstream sampler diverges at 3D scale (Sec. IV C).

F. Why benchmark numbers look worse than the literature's — protocol, not architecture

Readers who compare our tables and galleries against published reconstruction papers — including papers evaluated on the very same Kolmogorov data[8] — will notice that every method here, including the strong ones, scores and *looks* worse. We consider this observation a central result of the benchmark rather than an embarrassment, because each contribution to the gap is a protocol choice, and we measure them separately.

(i) Splits — severe in 3D, and, we were surprised to find, negligible in 2D. On the 617 consecutive DNS frames of the 3D cutout, frame-to-frame correlation is 1.00 at unit separation and 0.67 at separation 100, and a shuffled random-in-time split places every evaluation frame within one frame of a training frame. Under that split the latent flow-matching baseline scores 0.14 relative L_2; under the spatially disjoint protocol the same architecture, data, and budget scores 0.56, and the fleet shifts by factors of 2–4 *unevenly*, so leakage there does not merely inflate scores, it reorders methods.

We expected the 2D benchmark to reproduce this and it does not. We re-evaluated frozen checkpoints on frames drawn from *training* trajectories, positioned two raw frames from a training frame, where the measured field correlation to the nearest training frame is $r = 0.97$ (raw-frame correlation on this dataset: 0.992 / 0.972 / 0.916 at lags 1 / 2 / 4). Despite near-duplicate evaluation targets, no method gained materially: DMF-Gen 0.487 $\rightarrow$ 0.483, SiT 0.494 $\rightarrow$ 0.491, Senseiver 0.913 $\rightarrow$ 0.873, against a training-free inverse-distance control that moved 0.546 $\rightarrow$ 0.537 and therefore sets a $\approx$1.6% frame-difficulty floor. Net of that control, the largest memorization benefit in the fleet is under three percent.

The distinguishing variable is plausibly data scale rather than temporal proximity: the 2D training set holds 2560 snapshots against the 3D protocol's 150, and at that ratio these $\sim$6.5M-parameter models apparently learn the sensing operator rather than a lookup table, so a near-duplicate target buys them almost nothing. We report this as a measured contrast with a hypothesis attached, not a demonstrated mechanism. Its practical consequence is the useful part, and it is a caution against the easy version of our own argument: *leaky splits are not automatically fatal — they are fatal in the small-data regime where the 3D sparse-reconstruction literature actually operates*, which is precisely where random-in-time splits remain standard. The pre-fix 3D evaluations we archived were run on a single snapshot at a different solver setting, so we quote the latent-flow-matching figure that is directly comparable and do not tabulate the remainder.

(ii) What "1% sparsity" means. Much of the visually striking literature conditions on uniformly downsampled coarse grids — every cell of a coarse lattice, i.e. band-limited full-field information with bounded gaps — or adds physics guidance (PDE-residual or forcing terms at sampling time[8,10]). Our protocol uses randomly scattered points and purely data-driven conditioning: the same nominal percentage carries far less information and admits large sensor-free regions. Measured on Kolmogorov with the same checkpoint and frames, rearranging the 655 scattered sensors onto a uniform 25×26 lattice improves DMF-Gen from 0.487 to 0.367 relative L_2 (CRPS 0.235$\rightarrow$0.171) and IDW from 0.546 to 0.452 — a 25% accuracy difference from sensor *layout* alone, before any change of model or split. Layout dominates: combining the uniform lattice *and* the seen-trajectory split gives 0.364, statistically indistinguishable from the lattice alone, so at 2D data scales essentially the whole protocol gap that we can reproduce is sensor geometry rather than leakage.

(iii) What is shown. Publication figures typically display ensemble means or favorable frames; we display single posterior samples on fixed, pre-registered frames (our gallery frame runs 0.65–0.96 across the fleet against 0.39–0.50 fifty-frame averages), because single-sample behavior is what the physics metrics score. Ensemble size interacts with the same point: K=8 coverage understates calibration (Sec. II E), and ensemble means visually erase exactly the small scales the spectra measure.

(iv) Matched budgets. Every model trains at a matched $\sim$50k-step budget with disclosed tuning asymmetry, not to per-method convergence. Fairness costs polish; the cost table prices it.

The synthesis we intend a reader to take away: *apparent reconstruction quality in this* *literature is substantially a property of evaluation protocol, and only secondarily of architecture.* A benchmark that fixes the protocol first makes the remaining — genuine — architectural differences measurable.

VI. DISCUSSION

a. Which method should a practitioner use? The benchmark's answer is conditional, and that is the point. If sensing is dense relative to the flow's intrinsic dimension (laminar, periodic regimes) use gappy POD: on the cylinder it reaches 0.056 aggregate rel-L_2 against 0.093 for the best learned method, recovers the unobserved pressure channel, and costs a least-squares solve against a rank-20 basis. On observed channels of turbulence at $\sim$1% density, training-free interpolation remains competitive with the best learned methods — the case for learning begins where the problem is ill-posed: unobserved channels and sparser sensing. There, deterministic learners are structurally limited to conditional means; among generative routes, latent compression buys inference speed and global field structure at a measured dissipation-range floor, while ambient point-cloud generation buys discretization freedom and small-scale fidelity at higher inference cost. If the discretization exceeds $\sim 300^3$ or is body-fitted rather than Cartesian, grid families exit on measured walls or interface grounds. Uncertainty from any current method must be recalibrated before operational use, and the benchmark ships the wrapper.

b. Limitations. The three regimes are canonical and Cartesian: all are simulation data on periodic or lightly-bounded domains, so two axes the protocol is built to measure are specified but not exercised here — corrupted measurement operators (noise, occlusion, channel dropout) and surface-confined sensing on body-fitted unstructured meshes. Both are the subject of follow-up work on multiphysics LES and external-aerodynamics meshes; the capability column of Table II should be read as covering the regimes reported, not as a general claim. All learned rows train per-regime (no pretraining across flows); the suite contains no experimental data — particle-image-velocimetry or pressure-sensitive-paint regimes with real sensor noise are the natural next addition; single-snapshot reconstruction ignores temporal information that assimilation exploits; the cost accounting is single-GPU (H100) and does not probe multi-GPU scaling; the tuning-budget asymmetry of Sec. II F item (v) is disclosed but not eliminated; and training-seed variance is measured for only part of the fleet. [TODO: seed-replicate numbers when the replicate chains finish.]

VII. CONCLUSION

FlowRecon-Bench turns sparse-sensor flow reconstruction from a collection of incompatible per-paper evaluations into a measured landscape: three regimes spanning 2D laminar to 3D chaotic flow, twelve-plus audited methods, proper probabilistic scoring, physics metrics with their estimator pitfalls documented, matched cost accounting, and leakage-controlled splits with the distortion of the dominant protocol quantified. The findings that outlive any single table: method ranking is regime-dependent; interpolation is the floor to beat and sometimes is not beaten; unobserved channels of 3D turbulence are near-unidentifiable for every method; mean-field error alone misranks generative methods; every scored ensemble is under-dispersed and post-hoc recalibration repairs stated coverage; and the computational axis separates grid from point families at measured walls. The datasets, splits, sensor manifests, configurations, audit notes, and evaluation driver are released for extension — the leaderboard is meant to be beaten.

ACKNOWLEDGMENTS

[TODO: funding; compute; DMF-Gen authors; JHTDB providers.]

DATA AVAILABILITY

The benchmark artifact — datasets or regeneration recipes (JHTDB cutout coordinates and download scripts, OpenFOAM case definitions, converter scripts), split manifests, seeded sensor draws, per-method configurations, audit notes, and the evaluation driver — will be released on acceptance. [TODO: Zenodo/HF DOI; confirm JHTDB redistribution terms (fallback: recipe + coordinates); needs sign-off before anything goes public.]

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